Paleomagnetism of the Late Ordovician Juniata Formation and the Remagnetization Hypothesis

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The paleomagnetism of 103 samples from purplish gray sandstones of the Juniata Formation has been investigated. The samples were collected at 17 sites in Virginia, West Virginia, and Pennsylvania. Thermal demagnetization showed a strong magnetic overprinting in a large number of samples. This magnetization is secondary, since it was acquired during or after the late Paleozoic folding of the area; the resulting pole position, at 47°N, 129°E, agrees with published Late Carboniferous to Early Permian pole positions for North America. In all but 12 samples, directions of another component of magnetization could be determined in thermal demagnetization up to 672°C. These are assumed to represent the original magnetization of the Juniata Formation; they were acquired before the folding, and they show both normal and reversed polarities. The resulting pole position, at 31.5°N, 114.4°E (k = 55, α_{95} = 5°), falls directly in between published Middle Ordovician and Late Silurian pole positions for cratonic North America.

INTRODUCTION

Paleomagnetic studies of early and middle Paleozoic rocks can provide useful information about the history of an ancient ocean ('Iapetus') thought to have existed between North America and Europe [*Wilson*, 1966; *McKerrow and Ziegler*, 1972]. To date, however, the North American data are sparse and not always very reliable. The problems with the acquisition of new data are manifold, but one of the most serious is posed by remagnetization of the rocks involved.

Remagnetization was originally hypothesized by *Creer* [1968] as occurring at near equatorial latitudes and involving oxidation or possibly lateritization of relatively near surface rocks. For various geological and paleomagnetic reasons, Creer's hypothesis has since been rejected [*McElhinny*, 1973; *McElhinny and Opdyke*, 1973]. In a broader sense, however, remagnetization may also include thermal and chemical events in buried rocks [*Vincenz et al.*, 1975; *Van der Voo et al.*, 1977]. Since the latter effects may be most pronounced in those rocks that are now exposed in older orogenic belts, one may be inclined to avoid these areas in the search for suitable formations. Paradoxically, however, this eliminates at the same time potential usage of one of the best tests for documenting the relative age of a magnetization: the fold test of *Graham* [1949]. Moreover, other reasons prevail in selecting early Paleozoic sites for paleomagnetism near or in fold belts: more suitable rock types such as red beds, exposures of thicker sections, and better exposures and absence of glacial drift.

Previous paleomagnetic results from the Appalachian Paleozoic red bed sequences, including the Juniata, were mostly not obtained by using demagnetization techniques; these results are now generally discarded [*Graham*, 1949; *Howell et al.*, 1958; *Collinson and Runcorn*, 1960]. Exceptions are the studies of *Irving and Opdyke* [1965] and *Roy et al.* [1967], who investigated the Late Silurian Bloomsburg Formation. Two other more recent studies, of the Middle Ordovician Trenton limestone and the Beemerville alkalic intrusive complex [*McElhinny and Opdyke*, 1973; *Proko and Hargraves*, 1973], resulted in pole positions that fall near to that of the Bloomsburg Formation. Since areas such as Newfoundland and coastal New Brunswick conceivably did not form part of the continent [*McKerrow and Ziegler*, 1972], poles from these areas should not a priori be included in the apparent polar wander path of North America. Thus the three pole positions mentioned above are essentially the only ones from stable North America that are available to document the extent and closing history of the Iapetus Ocean. This study is intended to augment this data base. We must add, however, that recent

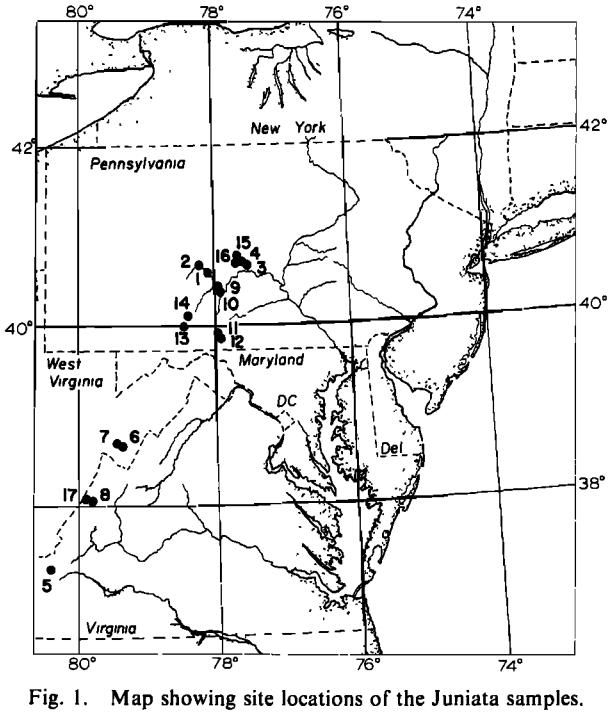


Fig. 1. Map showing site locations of the Juniata samples.

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TABLE 1. Summary of the Site Mean Directions of Magnetization and Their Statistical Parameters

Site	Location		Characteristic Magnetization, deg							Pole Position		
			Before Tilt Correction		After Tilt Correction							
	Latitude	Longitude	Declination	Inclination	Declination	Inclination	R_s	k	α_{95}, deg	Latitude, deg N	Longitude, deg E	R_r
1	40°33'55"	−78°12'55"	160.9	+38.5	156.7	+28.3	8/8	24.1	11.4	30.2	128.0	8/0
2	40°25'00"	−78°13'40"	12.1	+39.8	350.6	−27.7	4/6	21.1	19.8	34.1	112.6	1/3
3	40°39'20"	−77°35'05"	26.6	−78.7	338.7	−34.6	6/6	38.0	10.9	27.0	125.0	0/6
4	40°39'30"	−77°35'10"	69.5	−66.2	4.4	−42.5	7/7	38.4	9.8	24.5	98.0	0/7
5	37°17'15"	−80°28'50"	51.5	−71.2	341.9	−29.5	5/8	5.7	34.9	31.7	122.5	1/4
6	38°43'48"	−79°25'12"	170.7	+12.2	170.2	+23.1	8/8	16.1	14.2	38.3	112.7	3/5
7	38°43'48"	−79°25'48"	4.7	−48.0	350.8	−38.8	6/6	44.9	10.0	28.7	110.2	0/6
8	38°03'10"	−79°47'40"	170.0	−20.2	175.3	+30.1	4/4	283.3	5.4	35.5	105.8	4/0
9	40°23'30"	−77°54'55"	192.7	+43.4	172.5	+26.9	6/7	6.0	29.7	34.8	110.9	3/3
10	40°23'25"	−77°54'25"	11.5	−61.9	340.5	−46.6	3/4	12.1	37.0	19.3	120.2	0/3
11	39°56'05"	−77°57'50"	349.8	−21.9	342.7	−30.8	6/8	10.8	21.2	31.2	121.4	0/6
12	39°54'15"	−77°58'00"	3.0	−43.2	349.2	−36.6	6/6	6.2	29.1	28.8	113.5	0/6
13	40°00'50"	−78°28'35"	204.0	−32.1	193.4	+21.9	5/5	29.6	14.2	37.1	84.8	5/0
14	40°09'38"	−78°21'55"	204.4	+44.5	153.7	+31.5	2/3	...	...	27.7	130.1	1/1
15	40°46'55"	−77°37'15"	158.7	−2.9	163.6	+33.4	6/6	8.9	23.7	29.0	120.1	2/4
16	40°37'45"	−77°37'05"	185.9	−56.5	172.8	+33.8	7/7	26.3	11.9	30.3	110.2	7/0
17	38°03'10"	−79°47'40"	358.9	+35.9	349.1	−18.4	2/4	...	...	41.3	114.5	1/1

Mean	Characteristic Magnetization, deg							Pole Position		
	Before Tilt Correction		After Tilt Correction							
	Declination	Inclination	Declination	Inclination	R_s	k	α_{95}, deg	Latitude, deg N	Longitude, deg E	R_r
All site means before unfolding	184.1	+23.4	...	...	17/17	3.2	23.4	...	...	...
All site means after unfolding*	...	...	168.6	+31.8	17/17	53.0	4.9	31.8	113.8	...
All samples	...	...	169.2	+32.1	91/103	12.5	4.3	31.6	113.1	36/55
All reversed samples	...	...	172.2	+27.3	36/103	13.2	6.8	...	...	...
All normal polarity samples	...	...	347.1	−35.2	55/103	12.6	5.6	...	...	...
17 pole positions of the site means	...	...	...	...	17/17	55.7	4.8	31.5	114.4	...
Secondary directions before unfolding†	161.3	−1.8	...	...	53/103	7.6	7.5	47.3	129.3	53/0
Secondary directions after unfolding	...	...	164.7	−4.5	53/103	4.1	10.9	49.8	125.1	...

R_s is the ratio of the samples used in the statistical analysis to the number of samples measured; k and α_{95} are the statistical parameters [*Fisher*, 1953]; R_r is the ratio of samples with reversed polarity to the number of samples with normal polarity.

*Values for *dp* and *dm*, the semiaxes of the confidence ellipse about the pole position [*McElhinny*, 1973], are 3.1° and 5.5°, respectively.

†Values for *dp* and *dm* are 3.8° and 7.5°, respectively.

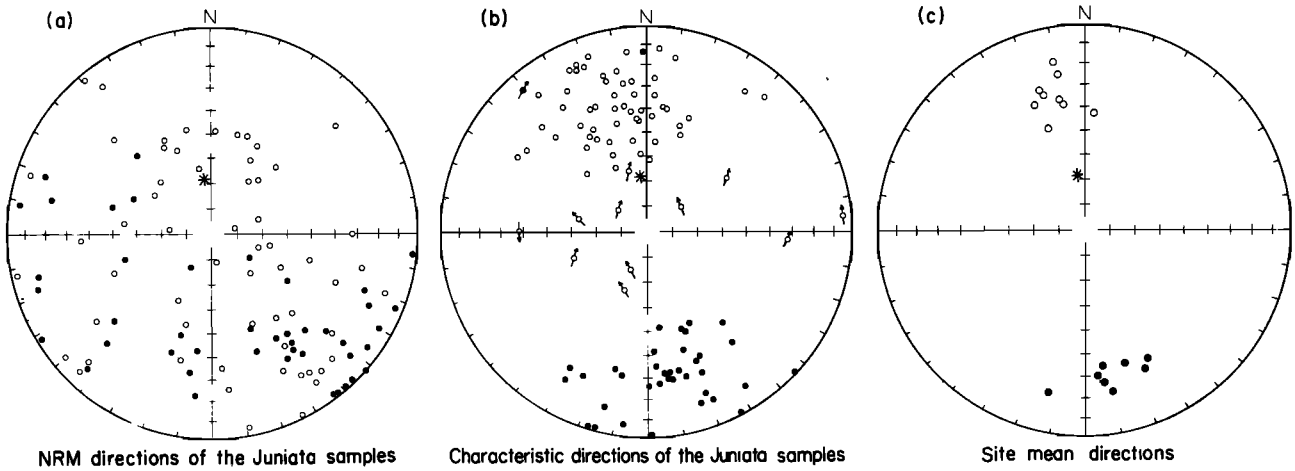


Fig. 2. Equal area projections of directions of magnetization, after correction for the tilt of the strata. Open (solid) circles represent projections on the upper (lower) hemisphere. Circles marked by arrows have not been used for the computations of the mean directions (see text). The asterisk is the present-day geomagnetic field direction.

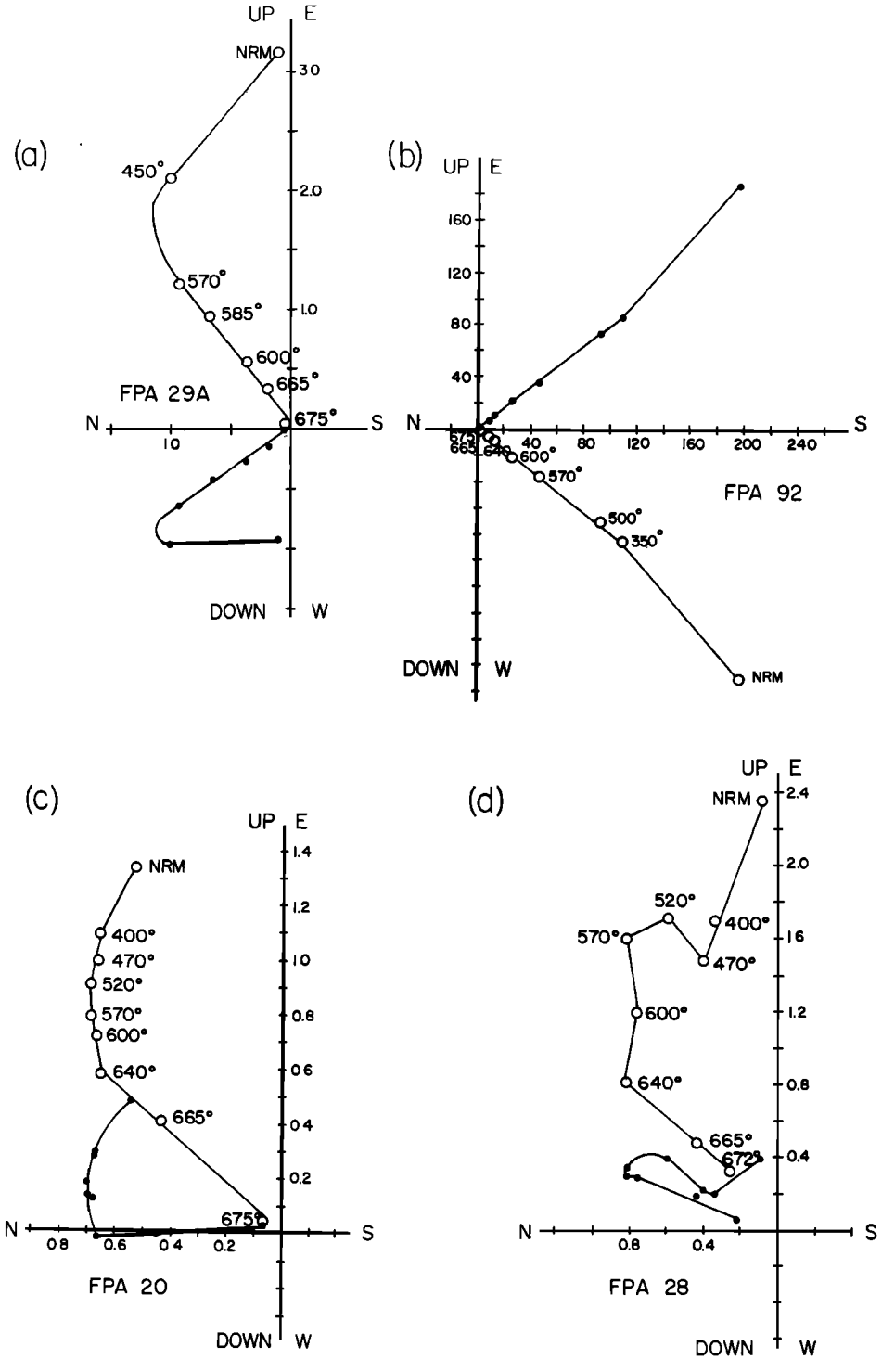


Fig. 3. Orthogonal demagnetization projections [*Zijderveld*, 1967] of the successive endpoints of the magnetization vector during progressive demagnetization, after correction for the tilt of the strata. Demagnetization temperatures are indicated in degrees Celsius. Solid circles represent projections on the horizontal plane, and open symbols those on the north-south vertical plane. Units indicated for the axes are in 10^{-3} A m^{-1} (10^{-6} emu/cm^3). Samples are from (*a*) site 4, (*b*) site 15, (*c*) site 3, and (*d*) site 4.

results from western Newfoundland [*Deutsch and Rao*, 1977] approximately match the results from stable North America. It may turn out that relative movements between western Newfoundland and stable North America have previously been overestimated.

Sampling and Measurements

Seventeen localities were sampled in Virginia, West Virginia, and Pennsylvania (Figure 1). Precise geographic locations are given in Table 1. The samples were generally taken

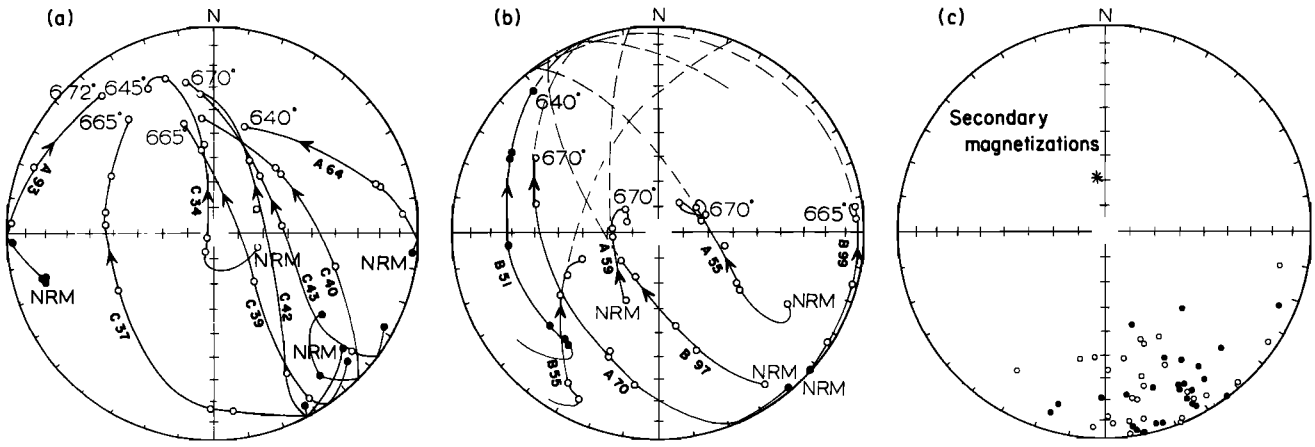


Fig. 4. Equal area projections of paths of the successive positions of the endpoints of the magnetization vector during thermal demagnetization, after correction for the tilt of the strata. Open (solid) circles represent projections on the upper (lower) hemisphere. Samples are from (*a*) sites 6, 7, 11, and 15 and (*b*) sites 5, 9, 10, 12, and 17. (*c*) Equal area projection of the directions of the intermediate components of magnetization, with respect to the present-day horizontal. Directions were obtained by vector subtraction of the magnetization vectors after demagnetization at about 400° and 600°C. The asterisk is the present-day geomagnetic field direction.

from purplish gray sandstones, though three sites consisted mostly of siltstones. At every site, attempts were made to have maximum stratigraphic coverage by obtaining drill cores or oriented hand samples from different beds. For the stratigraphy of the Late Ordovician Juniata Formation, we refer to *Thompson* [1970*a*].

Measurements were made with a Schonstedt spinner magnetometer (at the University of Michigan) or a Digico spinner magnetometer (at Lamont-Doherty Geological Observatory). The intensities of the natural remanent magnetizations generally ranged between 10^{-1} A m^{-1} (10^{-4} emu/cm^3) and 10^{-3} A m^{-1}. The directions were first referred to the present horizontal and then were corrected for tilting by rotation about the strike. Most of the sites were located in the long cylindrical anticlines of the valley and ridge province, and they were evenly distributed over the northwest and southeast flanks of the folds, in order to maximize the potential of the fold test and to minimize the rotational effects of simple tilt corrections within plunging folds.

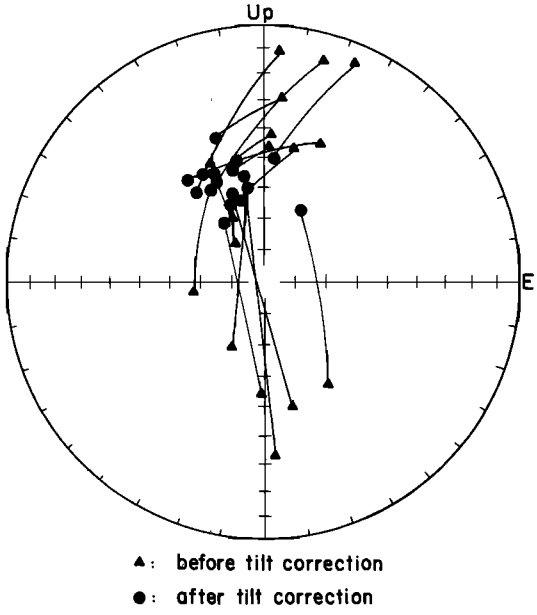


Fig. 5. Equal area projection of the 17 site mean directions, before and after correction for the tilt of the strata, showing *Graham*'s [1949] fold test applied to our samples. Note the unusual vertical projection through the east-west axis, chosen for clarity. Symbols represent projections on the northern hemisphere; all reversed site mean directions (Table 1) have been inverted to normal polarity before plotting.

Thermal demagnetizations were carried out routinely for all samples in at least four, but often six, steps, up to 672°C. Strong secondary magnetizations were present in almost all samples, resulting in an almost random distribution of natural remanent magnetization directions (Figure 2*a*). Upon demagnetization, however, most directions shifted considerably. In order to document the existence and directions of the secondary as well as the characteristic magnetizations we will describe the demagnetizations in some detail in the next section.

Demagnetization Analysis

The change in direction of the samples, upon demagnetization, was generally large, indicating a multivectorial natural remanent magnetization. In low temperatures, up to about 300°C, a magnetization component of small magnitude is removed. This component is assumed in most cases to be of recent origin, since it was directed steeply down close to the present-day field direction. Between temperatures of 400° and 600°C and often even higher, an intermediate component of magnetization is removed. The samples can be divided into three types, depending upon the magnitude and blocking temperatures of this intermediate component.

In the first type the intermediate component is not very important. Demagnetization diagrams [*Zijderveld*, 1967] for this set are shown in Figure 3. The intermediate component is absent (Figure 3*b*) or relatively small and removed between 400° and 570°C (Figure 3*a*). About 35% of the samples belong to this type.

In the samples of the second set the intermediate component is larger and has higher blocking temperatures (up to 640°C) as can be seen in Figures 3*c* and 3*d*. Above 640°C the magnetization vector appears to decay linearly to the origin. This high-temperature magnetization is the characteristic magnetization. About 20% of the samples behave in this manner.

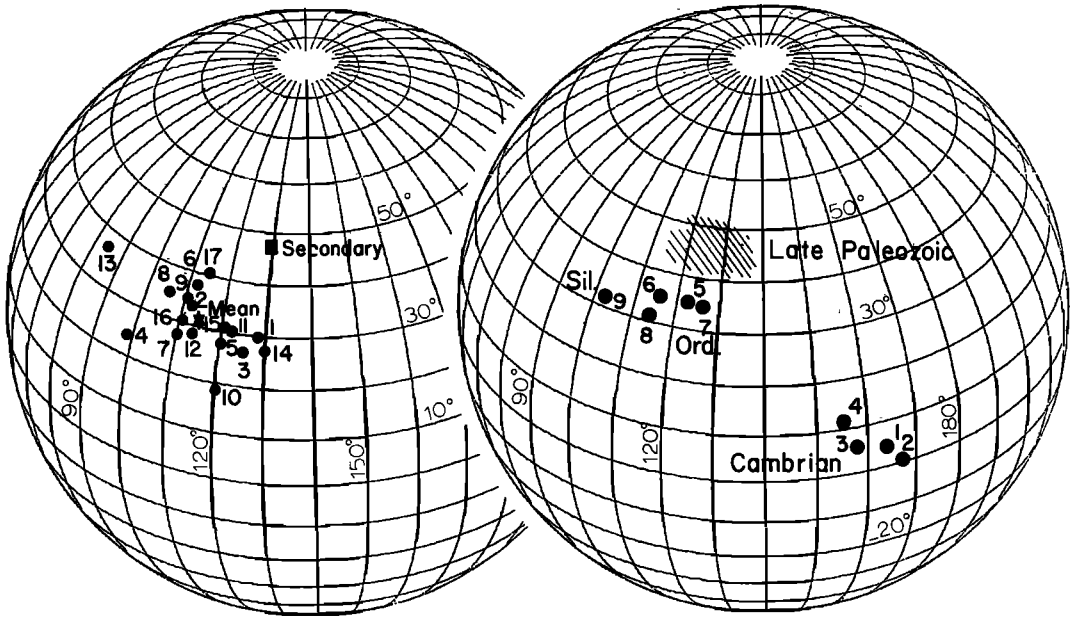


Fig. 6. Plot of pole positions relevant to this study. (Left) Site mean pole positions are shown, as well as their mean and the pole obtained from the secondary magnetizations. (Right) Our mean pole and published early Paleozoic pole positions are plotted for the stable part of North America. The numbers correspond to those in Table 2.

A third set of samples in which the intermediate component is very large possesses blocking temperatures as high as 670°C. Yet in all cases the blocking temperatures of the characteristic magnetizations are even higher, so that the directions keep shifting systematically during progressive demagnetization. In Figures 4*a* and 4*b*, equal area projections are shown illustrating this behavior. It can be seen that the paths during the removal of the intermediate component are converging to an upward north-northwestern direction, which is the characteristic direction of the other samples. These paths are essentially the great circles of remagnetization [*Halls*, 1976]. Since with this type a stable endpoint is not reached, the selection of the direction of characteristic magnetization is inherently subjective. For the samples of Figure 4*a* the directions chosen for statistical analysis are the endpoints of the paths. These directions may show more scatter than those of the other types, but they seem to be fairly evenly distributed about the extrapolated intersection points of the great circles of remagnetization. However, for some samples the endpoints of the paths were still far removed from the other characteristic directions (e.g., Figure 4*b*); these 12 samples are marked by arrows in Figure 2*b*, indicating their sense of directional change. They have not been used for the computation of the site mean directions (Figure 2*c* and Table 1), since it is considered that the samples still possess significant secondary components even at temperatures near the Néel point.

Particularly for the last set of samples (about 45% of the total) the directions of the intermediate components could be fairly well determined by vector subtraction. Generally, the trajectories between 400° and 600°C were used to calculate these directions, which are shown with respect to the present horizontal in Figure 4*c*. The application of the correction for the tilt of the strata causes a marked increase in the scatter of these directions (Table 1): k decreases from 7.6 to 4.1, and α_{95} increases from 7.5° to 10.9° (N = 53). Thus we conclude that these intermediate components were acquired during or after the Appalachian folding.

On the other hand, the characteristic directions show a significant improvement in their clustering upon unfolding (Figure 5). The corresponding statistical parameters are given in Table 1.

THE CHARACTERISTIC MAGNETIZATION: DISCUSSION

Table 1 summarizes the mean directions and their statistical parameters for each of the sites. Within-site precision is rather low for about half the sites, their precision parameters k falling below 25. On the other hand, the 17 site mean directions cluster well (Figure 2*c*), giving a mean direction with declina-

TABLE 2. Paleomagnetic Pole Positions (Plotted in Figure 6)

Site	Name	Geologic Age	Pole Position	Source
9	Bloomsburg Formation	Late Silurian	32.0N, 102.0E	[*Roy et al.*, 1967]
8	Juniata Formation	Late Ordovician	31.5N, 114.4E	this study
7	Beemerville alkaline complex	Late Ordovician	35.0N, 126.0E	[*Proko and Hargraves*, 1973]
6	Trenton limestone	Middle Ordovician	36.0N, 116.0E	[*McElhinny and Opdyke*, 1973]
5	Colorado intrusives	Cambro-Ordovician	36.6N, 122.3E	[*Larson and Mutschler*, 1971]
4	Colorado intrusives	Late Precambrian to Cambrian	12.0N, 156.0E	[*French et al.*, 1977]
3	Wilberns Formation	Late Cambrian	6.4N, 158.8E	[*Van der Voo et al.*, 1976]
2	Lamotte Formation	Late Cambrian	0.5N, 168.0E	[*Al-Khafaji and Vincenz*, 1971] thermal demagnetization
1	Wichita granites	Late Cambrian	4.0N, 164.0E	[*Spall*, 1970] thermal cleaning

tion 168.6°, inclination +31.8°, k = 53, and α_{95} = 4.9°. Predictably, it does not make a difference whether sample directions or site means are used in the computation of the overall mean direction (Table 1). The mean direction of all reversed samples is 172° removed from the mean of all normal polarity samples (Table 1); they are almost antipodal.

The pole position, obtained through averaging the pole positions of the 17 sites, falls at 31.5°N, 114.4°E. This is between the Late Silurian pole for the Bloomsburg Formation and the two Middle Ordovician poles for the Trenton limestone and the Beemerville complex (Figure 6 and Table 2).

This cluster of poles, now more firmly established, has been shown by *Morris* [1976] to be significantly different from pole positions from rocks in Great Britain. Morris showed that between Ordovician and Late Devonian time a significant north-south component of motion of North America, relative to Great Britain, must have occurred. Since paleolongitudes cannot be determined, it cannot be ascertained paleomagnetically whether this north-south motion is accompanied by a large east-west convergence between the continents (implying a large rate of subduction) or whether it is predominantly strike slip (implying hardly any subduction). Other evidence from geological and paleontological studies [e.g., *McKerrow and Cocks,* 1976] could be used but will not be discussed here, since this is beyond the scope of this paper.

THE SECONDARY MAGNETIZATION: DISCUSSION

In more than half of the samples the direction of a large secondary magnetization (Figure 4*c*) could be determined by vector subtraction, as described above. The mean directions before and after unfolding are not very different from each other, but the statistical parameters are significantly worse after unfolding, an indication that this component was acquired during or after the Appalachian folding. Moreover, its pole position agrees well with published Late Carboniferous and Early Permian poles for North America [*Van der Voo and French,* 1974].

Geochemical studies of the color genesis in the Juniata and underlying Bald Eagle formations [*Thompson,* 1970*b*] may offer some insight into the origin of the magnetizations. Thompson found that the iron oxides of the Juniata Formation were almost entirely hematites, and from the pigment between grains he concluded that at least part of this hematite was primary. On the other hand, secondary hematite was also observed. The Bald Eagle Formation, which consists of gray green siltstones and sandstones, is thought to have been originally reddish-colored as well [*Thompson,* 1970*b*, pp. 613–614]. Aqueous reduction and dissolution of the red pigment is his favored mechanism for this discoloration of the Bald Eagle Formation. It is conceivable that this chemical event was responsible for the secondary magnetizations, but this remains rather speculative. Anyhow, a thermal origin for the secondary magnetizations seems less likely. Since there is no evidence of substantial metamorphism in the area, and since the blocking temperatures of the secondary magnetizations are very high, theoretical studies [*Pullaiah et al.,* 1975] would tend to rule out a thermal cause (see also *Van der Voo et al.* [1977]).

Whatever the cause of the secondary magnetizations in the Juniata Formation as well as in other formations such as the Rose Hill Formation (discussed in a companion paper by *French and Van der Voo* [1977]), their inferred age of Late Carboniferous to Early Permian might indicate that they are related to the Appalachian orogeny. This suggestion merits further investigation.

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